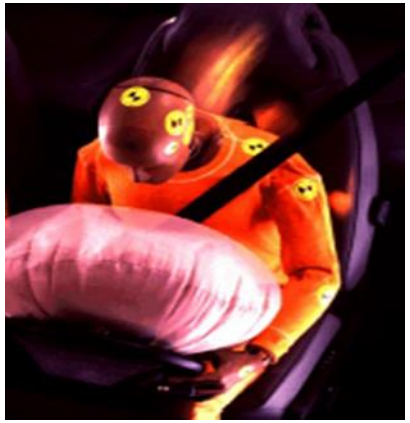
CHAPTER 2



Gaseous State

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The States of Matter

Matter: It is anything that has mass and occupies space.

Matter consists of very small particles called molecules:

- The molecules tend to fly apart because of their *Kinetic Energy.*
- There are attraction forces among molecules called *"Van der waal's forces"*
- Accordingly the matter could be gas, liquid or solid.

EXAMPLES : water

Bromine

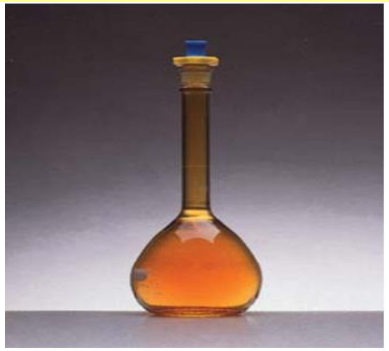
Solid liquid gas

To completely describe the state of a gaseous substance, 4 quantities are needed: volume, amount, temperature, and pressure.

1- Volume, V (Liters, cm^3, ...):

A gas expands uniformly to fill any container in which it is placed.

i.e. volume of a gas = volume of its container



NO_2 gas.

2- Amount:

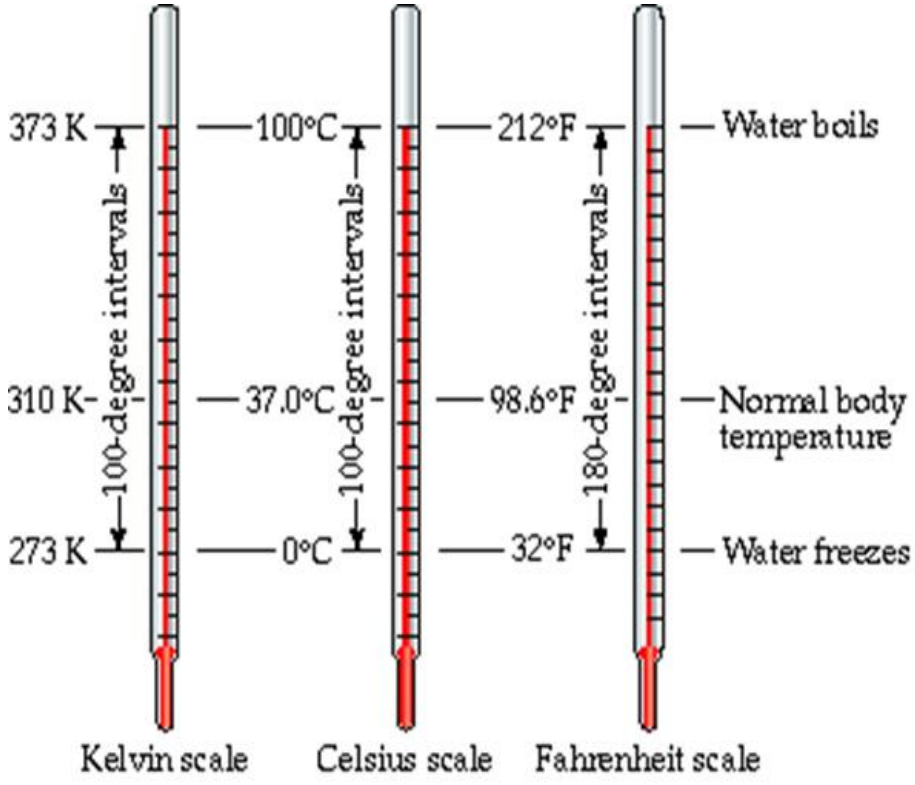
- **The amount of a gas is expressed in terms of: the number of moles (*n*) & the mass in grams (*m*).**

$$m = \mathcal{M} \times n$$

molar mass

3- Temperature:

$$T\,^{\circ}_{K} = t_{\circ C} + 273.15$$



4- Pressure; (force per unit area):

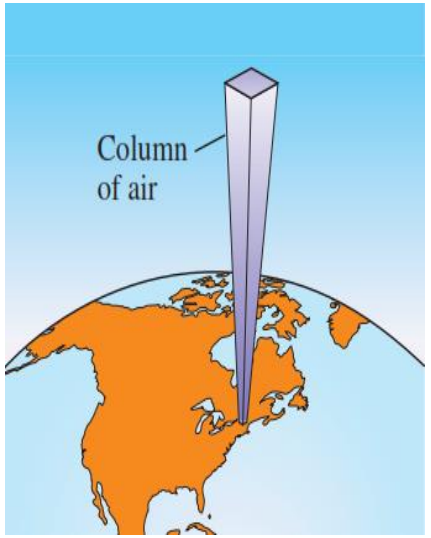
- **Pressures of gasses can be expressed in atm, or mmHg ,torr, Pascal, bar**

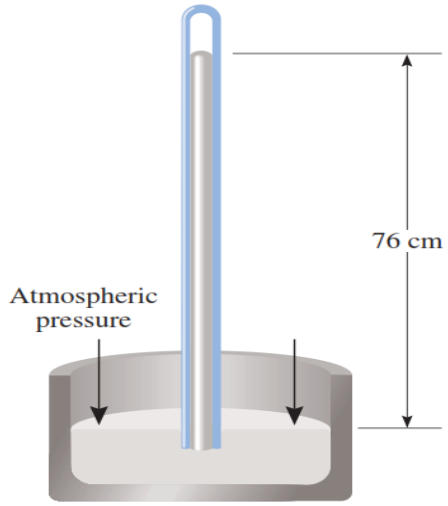
1 Pascal (Pa) = 1 N/m^2

1 atm = 760 mmHg = 760 torr

1 atm = 101,325 Pa = 101.325 kPa

1 atm = 1.01325 bar





The Ideal Gas Laws

1- Boyle's law:

"At constant temperature (T), the volume (V) of a fixed amount of a gas is inversely proportional to the pressure (P) on the gas."

$$V \propto \frac{1}{P} \quad \text{(at const. T)}$$

$$or\ V = \frac{K_1}{P} \quad ; (K_1 \text{ is proportionality constant})$$

$$Also; \qquad \boldsymbol{PV = K_1}$$

$$And \qquad P_1 V_1 = P_2 V_2 = K_1$$

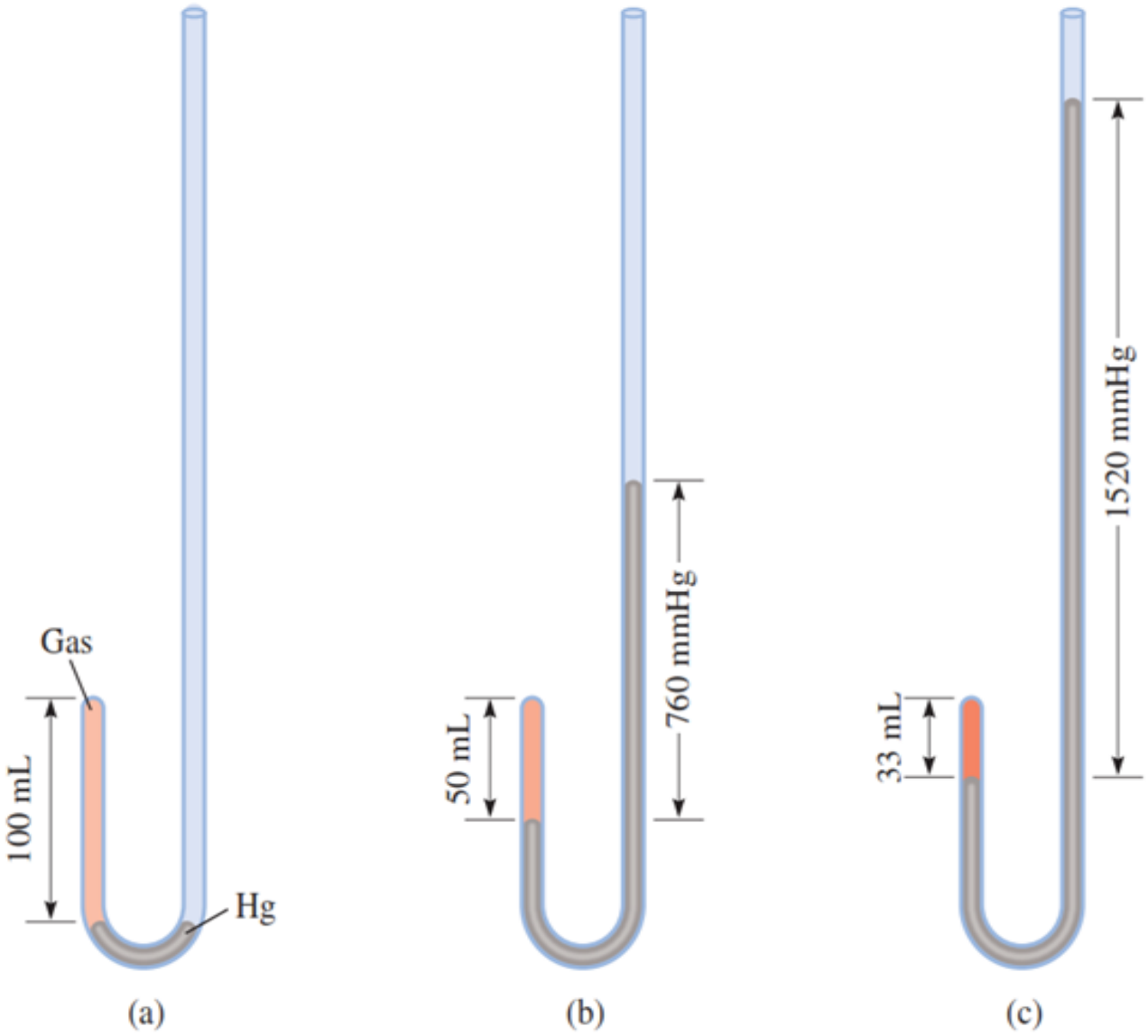
Thus if any three of these variables are known, the fourth one can be calculated.

$V \propto \frac{1}{P}$ (at const. T)

$PV = k_1$

TABLE 5.1 Actual Data from Boyle's Experiment

Volume (in^3)	Pressure (in Hg)	Pressure × Volume (in Hg × in^3)
117.5	12.0	14.1×10^2
87.2	16.0	14.0×10^2
70.7	20.0	14.1×10^2
58.8	24.0	14.1×10^2
44.2	32.0	14.1×10^2
35.3	40.0	14.1×10^2
29.1	48.0	14.0×10^2



Sample Exercise

Sulfur dioxide (SO_2), a gas that plays a central role in the formation of acid rain, is found in the exhaust of automobiles and power plants. Consider a 1.53-L sample of gaseous SO_2 at a pressure of 5.6×10^3 Pa. If the pressure is changed to 1.5×10^4 Pa at a constant temperature, what will be the new volume of the gas

Solution

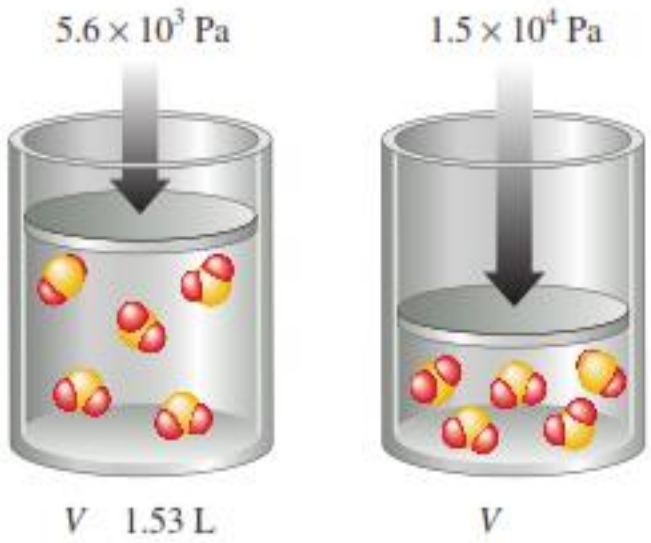
We can solve this problem using Boyle's law,

$$P_1V_1 = P_2V_2$$

$P_1 = 5.6 \times 10^3$ Pa $\quad P_2 = 1.5 \times 10^4$ Pa

$V_1 = 1.53$ L $\quad V_2 =$

$$V_2 = \frac{P_1V_1}{P_2} = \frac{5.6 \times 10^3 \text{ Pa} \times 1.53 \text{ L}}{1.5 \times 10^4 \text{ Pa}} = 0.57 \text{ L}$$



As pressure increases, the volume of SO_2 decreases.

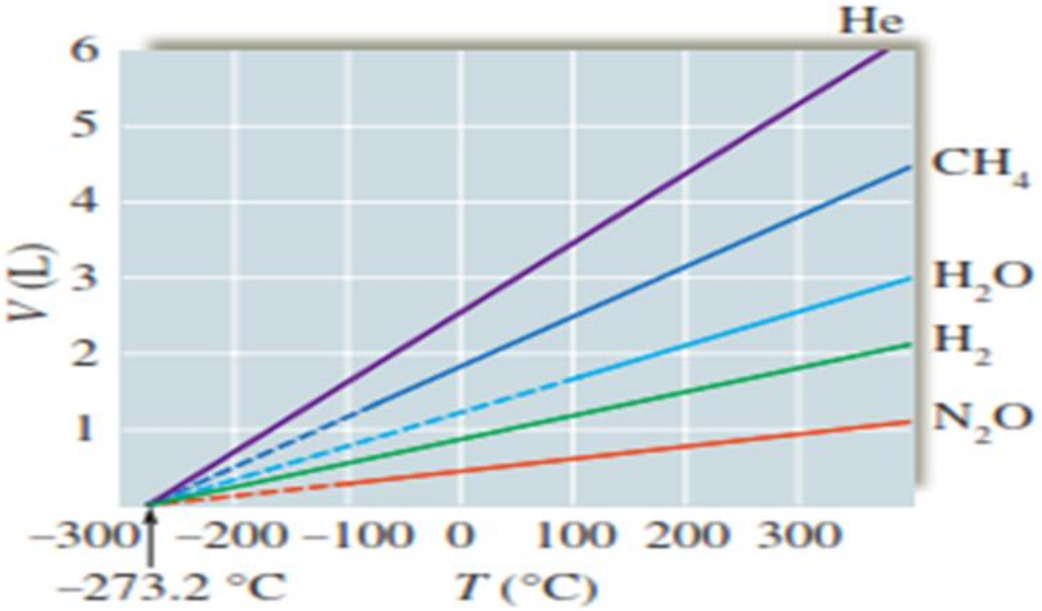
The new volume will be 0.57 L.

2- Charle's or Gay-Lussac's Law:

'At constant Pressure (P), the volume (V) of fixed amount of a gas is directly proportional to the absolute temperature (T)'.

- i.e. $V \propto T$ $(at\ constant\ P\&n)$
- Or $V = K_P T$ (where, K_P is the proportionality constant).

Or $$\frac{V_2}{V_1} = \frac{T_2}{T_1}$$



N.B. :The absolute or Kelvin temperature, $T_K = t_{°C} + 273.15$

Sample Exercise

A sample of gas at 15°C and 1 atm has a volume of 2.58 L. What volume will this gas occupy at 38°C and 1 atm

Solution

Charles's law, which describes the dependence of the volume of a gas on temperature at constant pressure, can be used to solve this problem.

Charles's law also can be written as

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$T_1 = 15\ C + 273 = 288\ K$ $\quad$ $T_2 = 38\ C + 273 = 311\ K$

$V_1 = 2.58\ L$ $\quad$ $V_2 =$

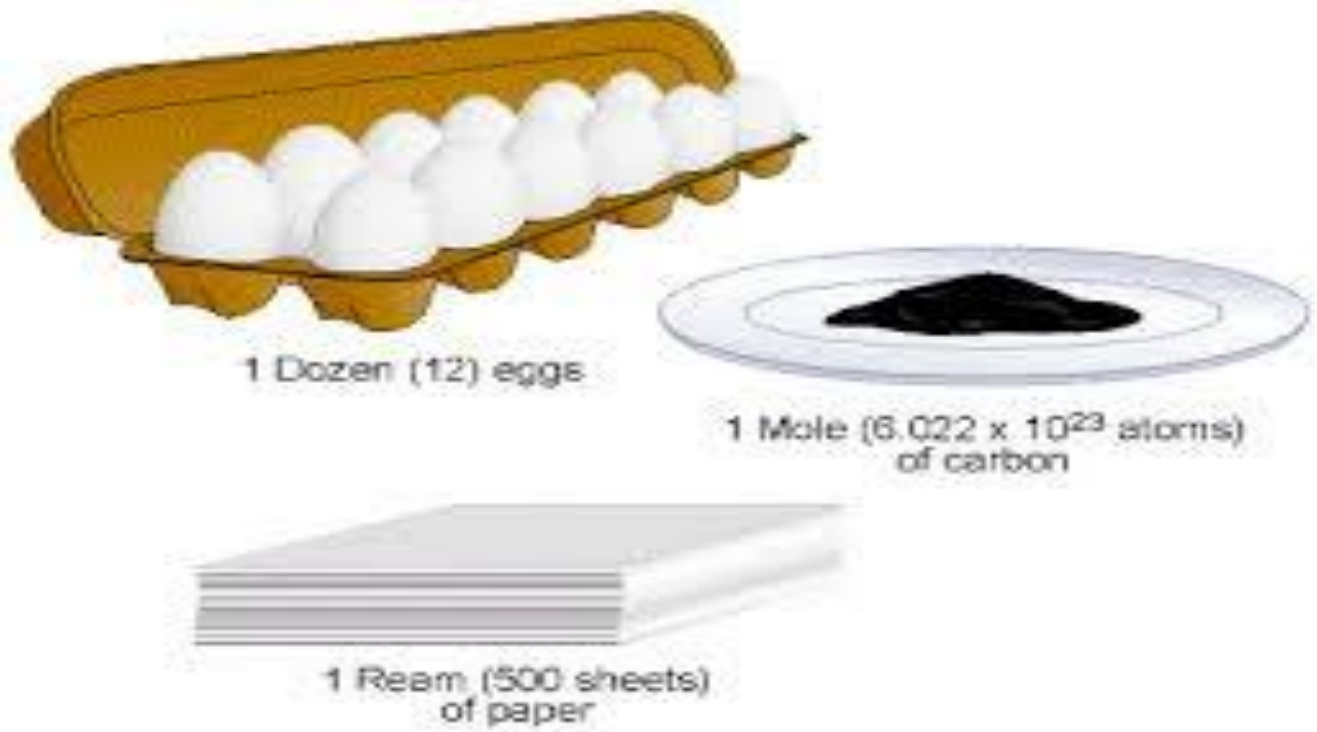
$$V_2 = \left(\frac{T_2}{T_1}\right) V_1 = \left(\frac{311\ K}{288\ K}\right) 2.58\ L = 2.79\ L$$

- **The Mole:**
- **Mole of a substance is its atomic mass or molecular mass expressed in grams.**

"one mole of any substance contains Avogadro's number of particles"

This number *(6.022×10^{23}) is called Avogadro's number.*

- **" One mole of a substance consists of 6.022×10^{23} units of that substance."**
- **The same as" one dozen of something contains 12 units of that thing."**



Element	Number of Atoms Present	Mass of Sample (g)
Aluminum	6.022×10^{23}	26.98
Copper	6.022×10^{23}	63.55
Iron	6.022×10^{23}	55.85
Sulfur	6.022×10^{23}	32.07
Iodine	6.022×10^{23}	126.9
Mercury	6.022×10^{23}	200.6

- ***Molar mass (molecular weight) ; M*** :

The molar mass of a substance is *the mass of one mole in grams.*

$$m = \mathcal{M} \times n$$

molar mass

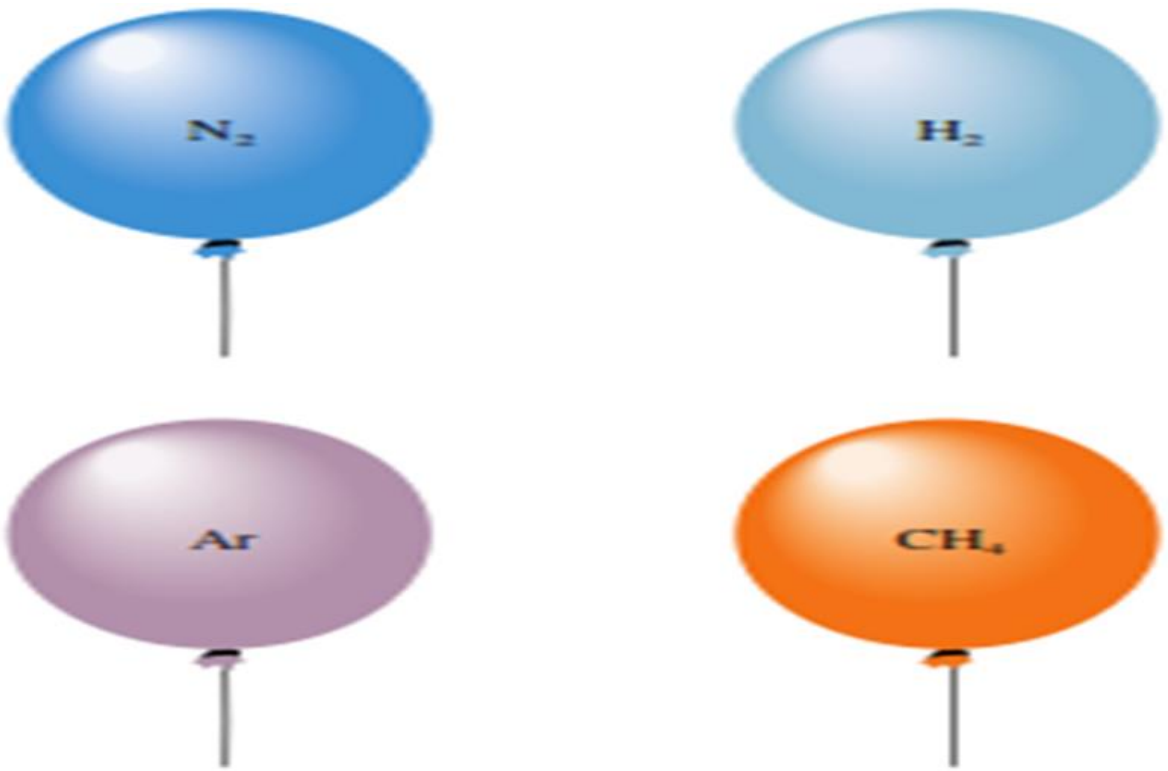
Avogadro's law:

"One mole of every gas under the standard conditions of temperature and pressure (STP) occupies the same volume (22.414 litres)"

- **STP: " It is a standard conditions at which T= 0 °C or (273.15 °K), and**
- **P=1 atm."**

- *Another form for Avogadro's law:*

"Equal volumes of gases at the same temperature and pressure contain the same number of particles".



These balloons each hold 1.0 L of gas at 25C and 1 atm. Each balloon contains 0.041 mol of gas, or 2.5x 10^{22} molecules.

Mathematically :

$$V \propto n$$

or

$$V = a\,n$$

Where:

V is the volume of the gas, n is the number of moles of gas particles, and a is a proportionality constant.

$3H_2(g)$	+	$N_2(g)$	$\longrightarrow$	$2NH_3(g)$
3 molecules	+	1 molecule	$\longrightarrow$	2 molecules
3 moles	+	1 mole	$\longrightarrow$	2 moles
3 volumes	+	1 volume	$\longrightarrow$	2 volumes

$$\underset{\text{3 volumes}}{3H_2(g)} + \underset{\text{1 volume}}{N_2(g)} \longrightarrow \underset{\text{2 volumes}}{2NH_3(g)}$$

The combined Gas law and ideal gas equation

The Pressure-Volume Relationship: Boyl's Law

$$V \alpha 1/P$$

The Temperature-Volume Relationship: Charles's Law

$$V \alpha T$$

The Amount-Volume Relationship: Avogadro's Law

$$V \alpha n$$

Combining the three Laws

$$V \alpha \frac{nT}{P}$$

$$V = R \frac{nT}{P}$$

$$PV = nRT$$

atm Liter mol Kilven

Ideal gas equation

R is general gas constant

Values of general (universal) gas constant (R):

Starting from the ideal gas equation: PV = nRT

- **for 1 mole of ideal gas at STP; (T = 0 ^oC = 273 ^oK, and P= 1 atm.):**

$$R = \frac{P\,V}{n\,T} = \frac{1\ atm\ \times 22.414\ L}{1\ mole\ \times 273.15\ ^oK}$$

$$= 0.082\ atm.L.mol^{-1}.^oK^{-1}$$

$$R = \frac{101.325\text{x}\ 10^3\ Pa\ \text{x}\ 22.414\ x\ 10^{-3} m^3}{1\ mole\ \text{x}\ 273.15^0 K}$$

$$= 8.31\ Pa.\ m^3.mol^{-1}\ .\ ^oK^{-1}$$

$$= 8.31\ J.\ mol^{-1}\ .\ ^oK^{-1}$$

Gas Law Calculations

Problem 1 (Final and initial states):

A 250-mL flask, open to the atmosphere, contains 0.0110 mol of air at 0°C.

On heating, part of the air escapes; how much remains in the flask at 100°C?

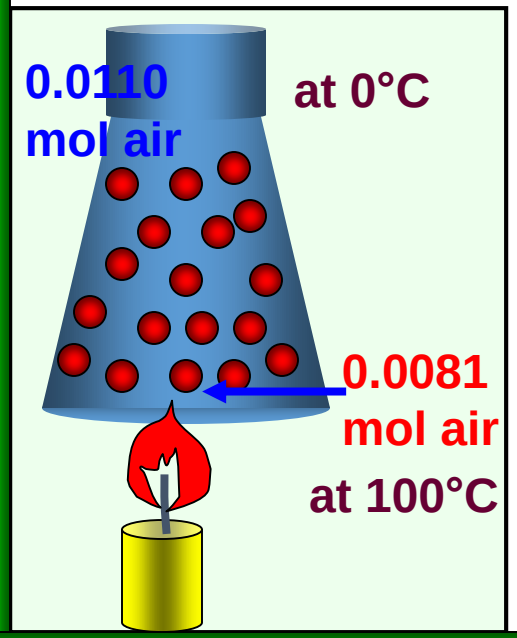
SOLUTION:

initial state:

$n_1 = 0.0110$ mol $\quad T_1 = 0 + 273 = 273$ K

$$P\ V = n_1\ R\ T_1$$

constant



$n_2 = ?$ $\quad T_2 = 100 + 273 = 373$ K

$$\frac{1}{P\ V} = \frac{x}{n_2\ R \times T_2}$$

constant

$$n_2 \times 373\text{K} = 0.0110 \text{ mol} \times 273\text{K}$$

$$n_2 = 0.008 \text{ mol}$$

Gas Law Calculations

Problem 2 (Calculation of P, V, n, or T) :

The ozone-friendly compound now used as a refrigerant in car air conditioners has the molecular formula $C_2F_4H_2$. If 2.50 g of this compound is introduced into an evacuated 500.0-mL container at 10°C , what pressure in atmospheres is developed?

SOLUTION:

$$P\ V = n\ R\ T$$

? → P

$\frac{500.0}{1000}$ L → V

$= \frac{m\ (g)}{\mathcal{M}\ (g/mol)} = \frac{2.50\ g}{102.04\ g/mol} = 0.0245\ mol$ → n

0.0821 L.atm/mol.K → R

10 + 273 = 283K → T

$\mathcal{M}\ (C_2F_4H_2)$ = (2 x 12.01) + (4 x 19) + (2 x 1.008) = 102.04

P = 1.14 atm

Gas Law Calculations

Problem 3 (Molar Mass):

A sample of liquid acetone is placed in a 300-mL flask and vaporized by heating to 95°C at 1.02 atm. The vapor filling the flask at this temperature and pressure weighs 0.587 g. Calculate the molar mass of acetone.

SOLUTION:

$$P\ V = n\ R\ T$$

1.02 atm

$\frac{300}{1000}$ L

0.0821 L.atm/mol.K

95 + 273 = 368K

$$= \frac{m\ (g)}{\mathcal{M}\ (g/mol)} = \frac{0.587\ g}{\mathcal{M}\ (g/mol)}$$

$$\mathcal{M} \approx 58.0\ g/mol$$

Argon is an inert gas used in light bulbs to retard the vaporization of the filament. A certain light bulb containing argon at 1.20 atm and 18 °C is heated to 85 °C at constant volume. What is the final pressure of argon in the light bulb (in atm)?

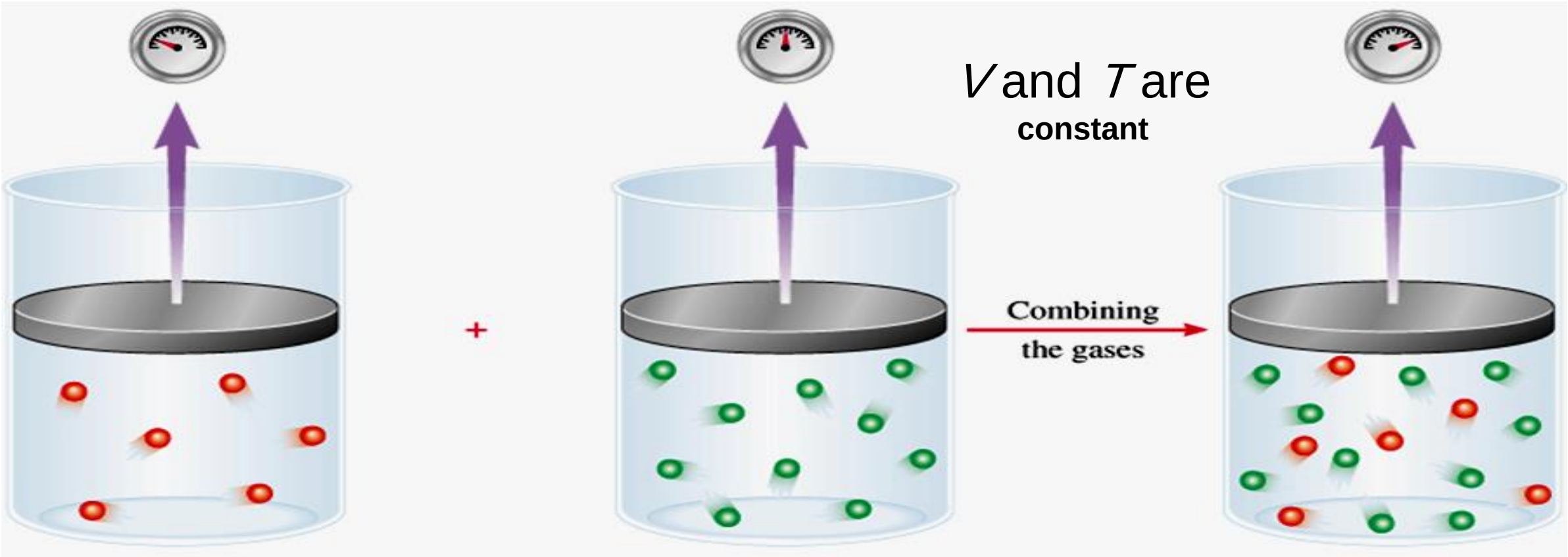
SOLUTION:

PV = nRT; (n, V and R are constant) $\therefore (P/T) = (nR/V)$

$\therefore (P_1/T_1)=(P_2/T_2)$

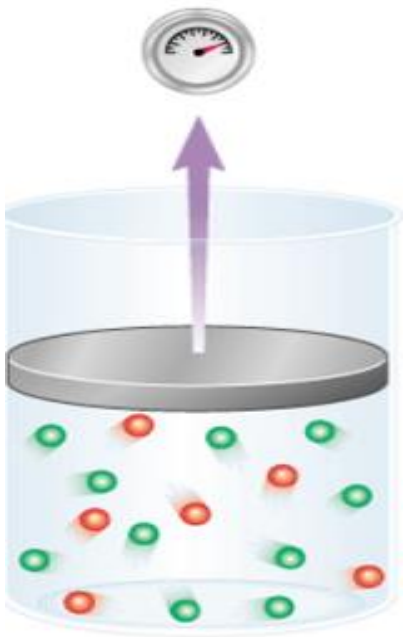
$\therefore$ $[P_2 = (P_1 x T_2/T_1) = (1.2 \text{ atm} \times 358K/291K)] = 1.48 \text{ atm}$

Dalton's Law of Partial Pressures (for mixture of gases)



P_1 P_2 $P_{total} = P_1 + P_2$

Consider a case in which two gases, A and B, are in a container of volume V.



$$P_A = \frac{n_A RT}{V}$$

n_A is the number of moles of A

$$P_B = \frac{n_B RT}{V}$$

n_B is the number of moles of B

$$P_T = P_A + P_B$$

$$X_A = \frac{n_A}{n_A + n_B} \qquad X_B = \frac{n_B}{n_A + n_B}$$

$$P_A = X_A P_T \qquad P_B = X_B P_T$$

$$\boxed{P_i = X_i P_T}$$

mole fraction (X_i) = $\frac{n_i}{n_T}$

A sample of natural gas contains 8.24 moles of CH_4, 0.421 moles of C_2H_6, and 0.116 moles of C_3H_8. If the total pressure of the gases is 1.37 atm, what is the partial pressure of propane (C_3H_8)?

$P_i = X_i P_T$ $\qquad$ $P_T = 1.37$ atm

$\therefore X_{propane} = [(0.116)/(8.24+0.421+0.116)]=0.0132$

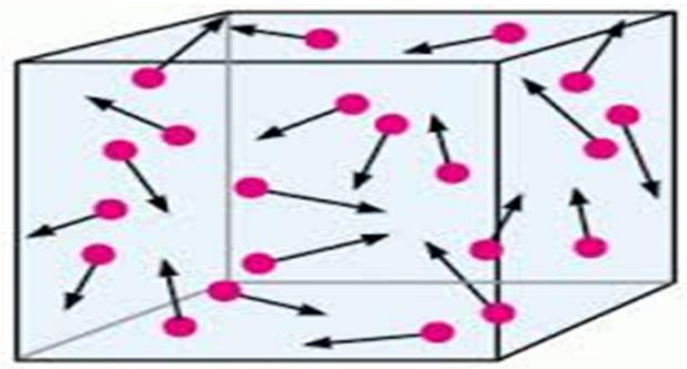
$P_{propane} = 0.0132 \times 1.37$ atm $= 0.0181$ atm

The Kinetic Molecular Theory of Ideal Gases

Postulates of the kinetic molecular theory describing an ideal gas:

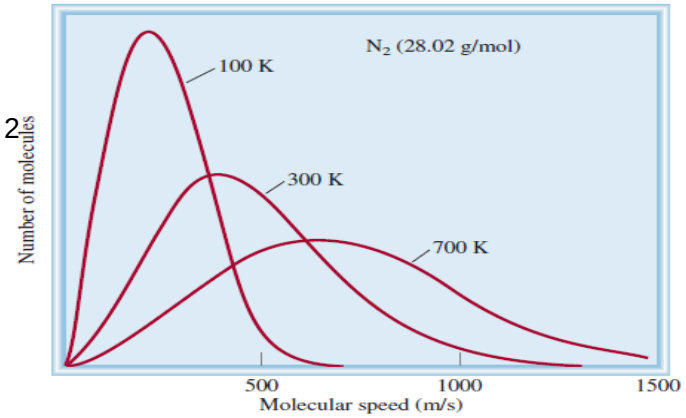
1. *The particles are so small compared with the distances between them;*
   i.e. ***Volume of the individual particles can be neglected (zero).***
2. *The particles are in constant rapid and random motion.*
3. *The collisions of the particles with the walls of the container are the cause of the pressure exerted by the gas.*
4. *The molecules are assumed to be perfectly elastic; (no net loss of energy).*
5. *The particles are assumed to exert no forces on each other;*
   No attraction or repulsion forces among ideal gas molecules.
6. *The average kinetic energy of the gas particles is directly proportional to the Kelvin temperature of the gas.*

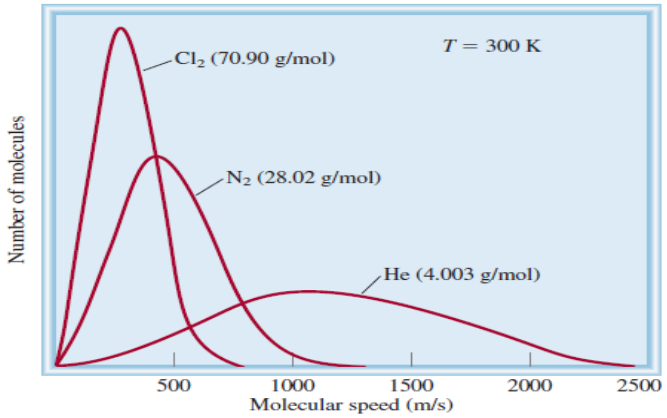
$$K.E. \propto T\ (^{o}K)$$



Root-mean- square speeds of gases

distribution of speeds for N_2 gas molecules
at 3 different temp





Distribution of speeds of 3 different gases at the same temperature

$$\sqrt{\overline{u^2}} = u_{rms} = \sqrt{\frac{3RT}{\mathcal{M}}}$$

Calculate the root-mean-square speeds of He atoms in m/s at 25°C?

To calculate u_{rms} , the units of R should be 8.314 J/K . mol and, because 1J = 1 kg m^2 /s^2 , the molar mass must be in kg/mol. The molar mass of He is 4.003 g/mol, or 4.003 x10^{-3} kg/mol

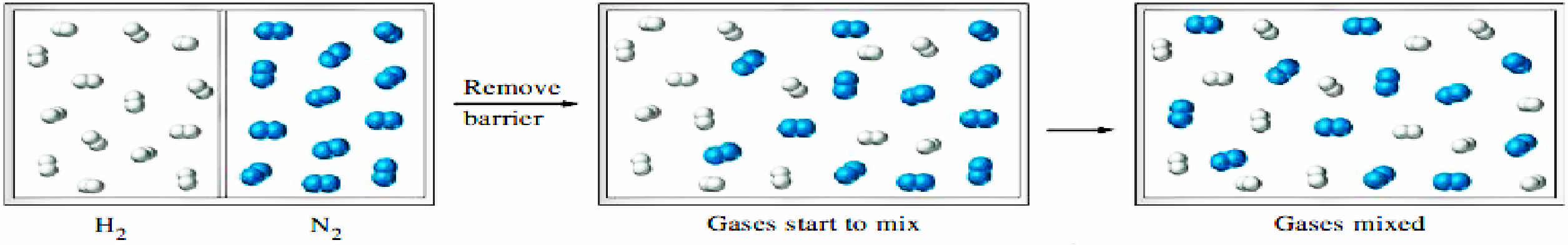
$$u_{rms} = \sqrt{\frac{3RT}{\mathcal{M}}} = \sqrt{\frac{3(8.314\ \text{J/K}\cdot\text{mol})(298\ \text{K})}{4.003\times10^{-3}\ \text{kg/mol}}} = \sqrt{1.86\times10^{6}\ \text{J/kg}}$$

Using the conversion factor 1 J = 1 kg m^2 /s^2 we get

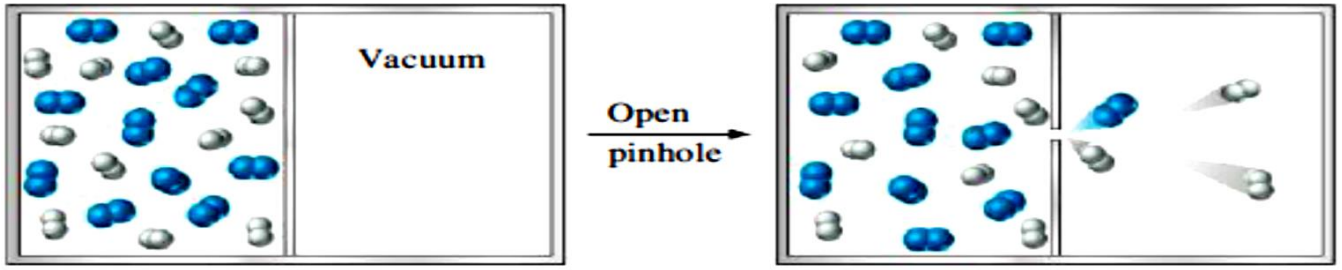
$$u_{rms} = \sqrt{1.86\times10^{6}\ \text{kg m}^2/\text{kg}\cdot\text{s}^2} = \sqrt{1.86\times10^{6}\ \text{m}^2/\text{s}^2} = 1.36\times10^{3}\ \text{m/s}$$

Gas Properties Relating to the Kinetic-Molecular Theory

Gas diffusion is the gradual mixing of molecules of one gas with molecules of another by virtue of their kinetic properties. The rate of diffusion is proportional to molecular speed. Diffusion generally proceeds from a region of higher concentration to a region of lower concentration.



Effusion: is the escape of gas molecules from their container through a tiny hole or porous membrane into vacuum



Graham's Law of Effusion

Graham proposed his law in 1831 to describe the diffusion of gases, but the law actually does not apply to diffusion. It applies only to (1) effusion, (2) for gases at constant low pressure (natural escape, not a jet), (3) escape from tiny hole without any collision. Molecules of a diffusing gas undergo collisions with each other and with the gas into which they are diffusing. Some even move in the opposite direction to the net flow. Nevertheless, diffusion does occur, and gases of low molar mass do diffuse faster than those of higher molar mass. We can't, however, use Graham's law to make quantitative predictions about rates of diffusion!

$$\frac{rate\,of\ effusion\,of\ A}{rate\,of\ effusion\,of\ B} = \frac{(u_{rms})_A}{(u_{rms})_B} = \sqrt{\frac{3RT/M_A}{3RT/M_B}} = \sqrt{\frac{M_B}{M_A}} = \sqrt{\frac{d_B}{d_A}}$$

- The ratio of molar masses or densities of effusing gases can be used to compare: effusion rates, molecular speeds, effusion times, distances travelled or amounts of gases effused! (**rate α speed α distance travelled α amount of gas diffused α 1/time**)

Certain volume of O_2 gas effuse through inlet in 30 min. Find the time required for the same volume of CO_2 to effuse at the same conditions?

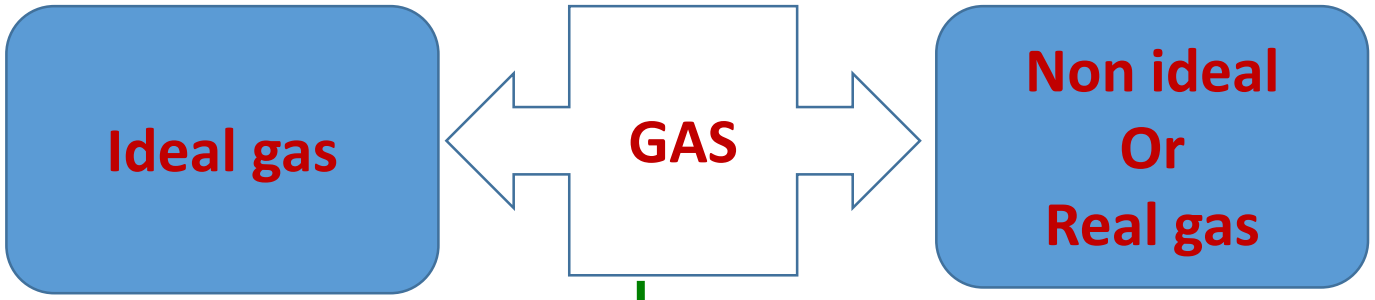
$$\frac{(rate\,of\ effusion)\,1}{(rate\,of\ effusion)\,2} = \frac{t_2}{t_1} = \sqrt{\frac{M_2}{M_1}}$$

The faster the diffusion rate the smaller the time of diffusion.

∴ Diffusion rate is inversely proportional to the diffusion time.

∴ $t_2 = t_1*(M_2/M_1)^{0.5} = 30*(44/32)^{0.5} = 35.2$ min

The Gaseous State



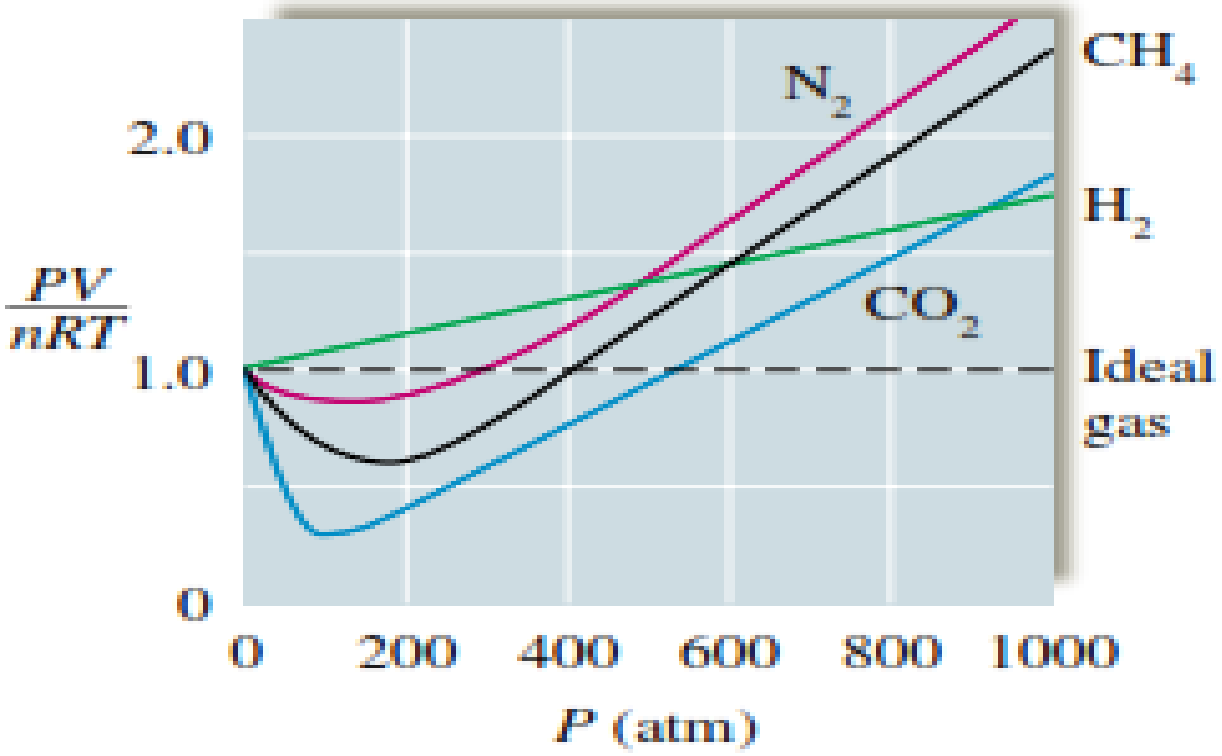
Ideal gas	Non ideal Or Real gas
1- obeys the gas laws such as *Charle's , and Boyle's law* at all temperatures (T) and pressures (P).	1- A real gas obeys the gas laws at low pressures and high temperatures.
2- Volume of the gas molecules is very small and can be neglected compared to the volume of the container.	2- Volume of gas molecules cann't be neglected.
3- There are no attraction or repulsion forces among the gas molecules.	3- the molecules of real gas exert some attractive or repulsive forces on each others.

Generally:

The gas behaves ideally when the <u>temperature is very high</u> and the <u>pressure is very low.</u>

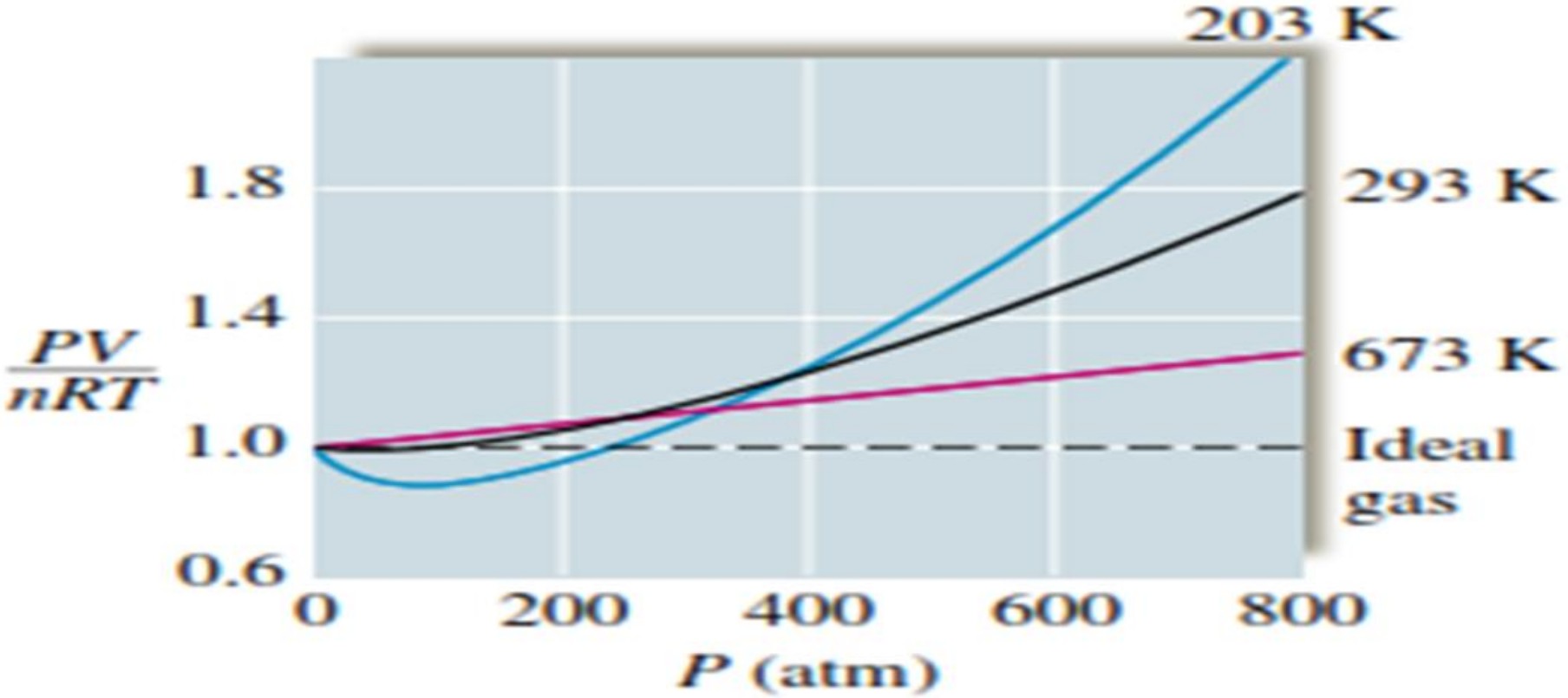
Real (Non-Ideal)Gases and the deviations from ideal behaviour

- The gas is said to be an *ideal gas* if it obeys the various gas laws, expressed in the form of *equation PV= nRT*.
- Very few gases obey the ideal gas equation only at low pressures and high temperatures.



Plots of ***PV/ nRT*** *versus* ***P*** *for several gases at* (200 °K).

- For ideal behavior ***(PV/ nRT = 1)***.
- *The* real gas is close to ideal behaviour only at low pressures (less than 1 atm).



*Plots of **PV/nRT** versus **P** for nitrogen gas at three temperatures.*

- **Note that although non-ideal behavior is evident in each case, the deviations are smaller at the higher temperatures.**

Causes of deviations of real gases from the ideal behavior:

- ***The KTG assumed that:***

1- the volume occupied by the gas molecules is very small and neglected compared to the container volume; and

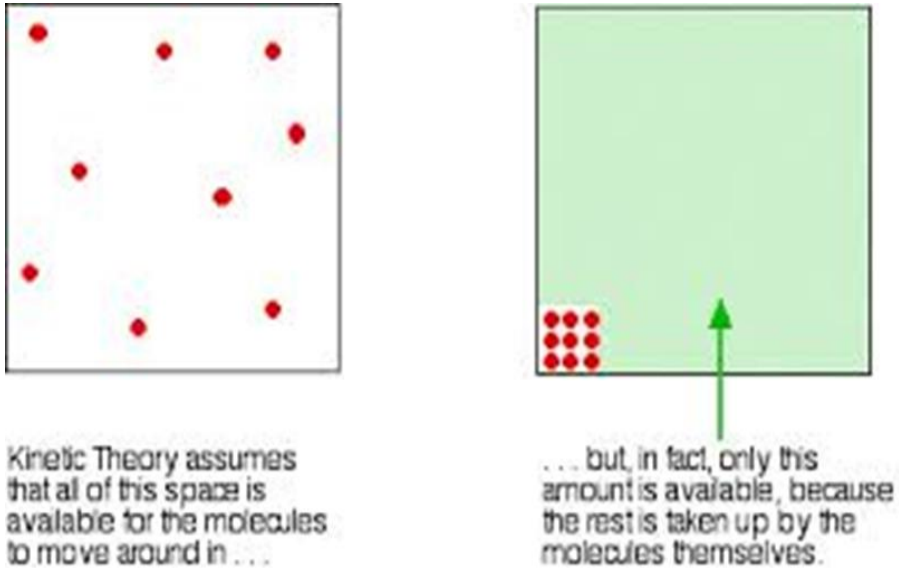
2-Gas molecules do not exert any force of attraction on each other.

In reality, both assumptions are wrong & need to be corrected to be applied to real gases .

- ***Modification of the Ideal Gas Equation: "Van der Waals Equation"***

1- Correction for the finite volume of the molecules:

- **In fact the space available for the gas particles to move in is less than the volume of the container because the gas molecules themselves occupy some volume.**



- *Van der Waals* suggested that the volume occupied by the molecules should be *subtracted* from the total volume of the gas in order to get the actual volume (ideal volume; V_i):

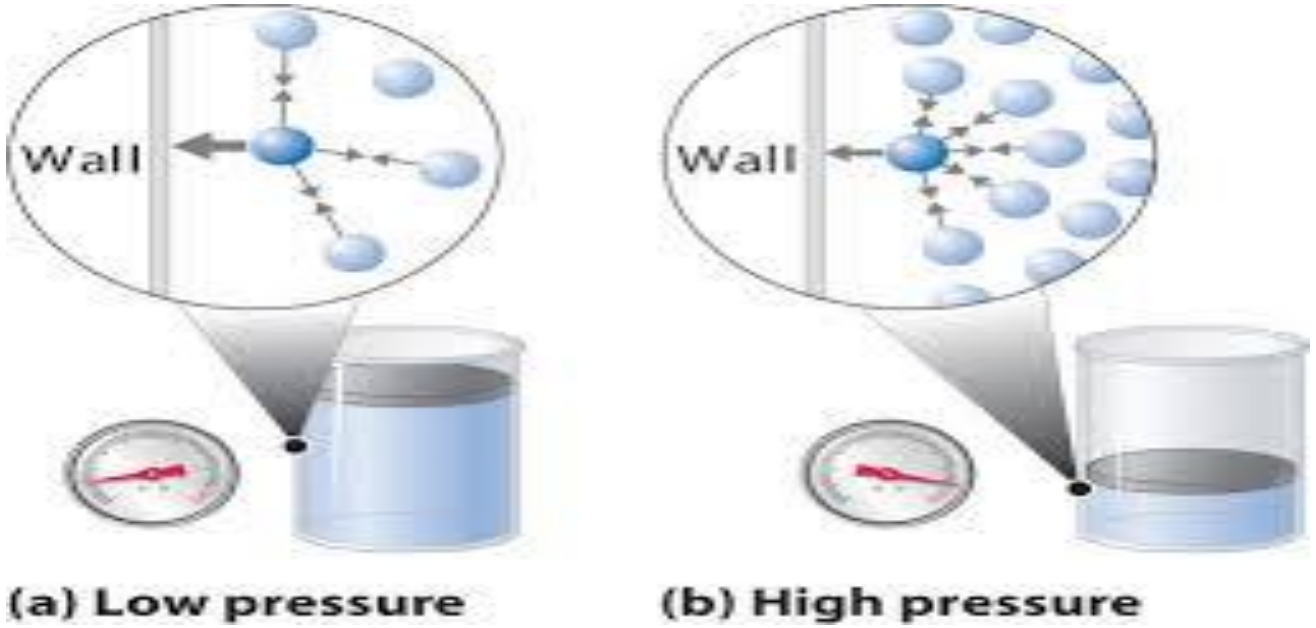
Thus:

$$V_i = V - nb$$

where: V is the volume of container,
n is the number of moles,
and b is the excluded volume for one mole.

2-correction for the attractive forces operating between the molecules:

In real gas; at high pressures the particles come close together and *attractive forces* between them cause the particles to hit the wall slightly less than they do in the absence of interactions.

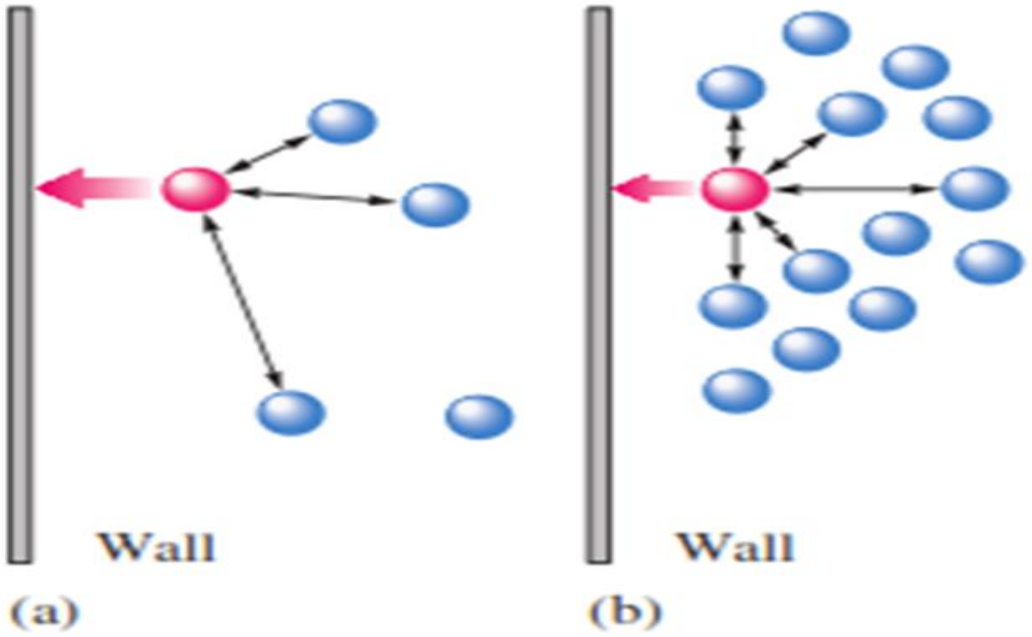


The observed pressure P_{obs} becomes smaller than that if the gas particles did not interact(P_i).

- Or $P_{obs} = P_i$ – correction factor

The correction factor is a measure of the attraction forces between molecules.

- It depends on the concentration of gas molecules in terms of $(n/V)^2$:



- *Thus* $$P_{obs} = P_i - a\left(\frac{n}{V}\right)^2$$

  where *a is a proportionality constant.*

Inserting the corrections for both the volume of the particles and the attractions of the particles gives the **Van der Waals equation:**

$$\underbrace{\left[P_{obs} + a\left(\frac{n}{V}\right)^2\right]}_{\substack{\text{Corrected pressure} \\ P_{ideal}}} \times \underbrace{(V - nb)}_{\substack{\text{Corrected volume} \\ V_{ideal}}} = nRT$$

The values of *a and b for various gases*

Gas	$a\left(\frac{atm \cdot L^2}{mol^2}\right)$	$b\left(\frac{L}{mol}\right)$
He	0.0341	0.0237
Ne	0.211	0.0171
Ar	1.35	0.0322
Kr	2.32	0.0398
Xe	4.19	0.0511
H_2	0.244	0.0266
N_2	1.39	0.0391
O_2	1.36	0.0318
Cl_2	6.49	0.0562
CO_2	3.59	0.0427
CH_4	2.25	0.0428
NH_3	4.17	0.0371
H_2O	5.46	0.0305

Practical exercise:

Given that 3.50 moles of NH_3 occupy 5.20 L at 47°C, calculate the pressure of the gas (in atm) using (a) the ideal gas equation and (b) the van der Waals equation.

For NH_3 gas , a= 4.17 **L^2atm/mol^2** b= 0.0371 **L/mol**

Answer's Key:

1. Applying ideal gas equation: **P= 17.7 atm.**
2. Applying Van der waal's equation: **P= 16.2 atm.**

Problems

1- A sample of gas occupies 12 L under a pressure of 1.2 atm. What would its volume be if the pressure were increased to 2.4 atm? [6L]

2- A sample of neon occupies 105 liters at 27°C under a pressure of 985 torr. What volume would it occupy at standard temperature and pressure (STP)? [124 L]

3- A sample of gas occupies 12.0 liters at 240.°C under a pressure of 80.0 kPa. At what temperature would the gas occupy 15.0 liters if the pressure were increased to 107 kPa? [858K, 585 °C]

4- A 0.109-gram sample of a pure gaseous compound occupies 112 mL at 100.°C and 750. torr. What is the molecular weight of the compound? [30.02 g/mol].

5- A 10.0-liter flask contains 0.200 mole of methane, 0.300 mole of hydrogen, and 0.400 mole of nitrogen at 25°C. (a) What is the pressure, in atmospheres, inside the flask? (b) What is the partial pressure of each component of the mixture of gases? [2.2 atm, 0.489, 0.734, 0.979]

6- Calculate the pressure exerted by 1.00 mole of methane, CH_4, in a 500.-mL vessel at 25.0°C assuming (a) ideal behavior and (b) nonideal behavior. (a= 2.25 L^2atm/mol^2 , b= 0.0428 L/mol) [48.9 atm, 44.6 atm]

7- A gas sample contained in a cylinder equipped with a moveable piston occupied 300. mL at a pressure of 2.00 atm. What would be the final pressure if the volume were increased to 500. mL at constant temperature?

8- A balloon that contains 1.50 liters of air at 1.00 atm is taken under water to a depth at which the pressure is 3.00 atm. Calculate the new volume of the balloon. Assume that the temperature remains constant.

9- Which of the following statements are true? Which are false? Why is each true or false? *Assume constant pressure* in each case.
(a) If a sample of gas is heated from 100°C to 200°C, the volume will double.
(b) If a sample of gas is heated from 0.°C to 273°C, the volume will double.
(c) If a sample of gas is cooled from 1273°C to 500.°C, the volume will decrease by a factor of 2.
(d) If a sample of gas is cooled from 1000.°C to 200.°C, the volume will decrease by a factor of 5.
(e) If a sample of gas is heated from 473°C to 1219°C, the volume will increase by a factor of 2.

10- 247-mL sample of a gas exerts a pressure of 3.13 atm at 16.0°C. What volume would it occupy at 100.°C and 1.00 atm?

11- A 280.-mL sample of neon exerts a pressure of 660. torr at 26°C. At what temperature in °C would it exert a pressure of 940. torr in a volume of 440. mL?

12- Find the value of the universal gas constant (*R) in L .atm/mol. K.*